SARDAR RAFAYET BIN MURTAZA

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EDUCATION

Michigan State University — B.S. in Electrical Engineering, East Lansing, MI • *Expected May 2027* • Dean's List: Fall 2023, Spring 2024, Spring 2025

Relevant Coursework: Electronic Circuits, Electronics Laboratory, Principles of Electronic Devices, Electromagnetic Fields & Waves I, Circuits & Systems I/II, Microprocessors & Digital Systems, Introduction to Signal Processing, Control Systems, Electrical Engineering Analysis, Probability & Statistics for Engineering

In Progress (Fall 2026): Microelectronic Fabrication, Energy Conversion & Power Electronics, Computer Aided Manufacturing

LEADERSHIP

Electrical Co-Lead — Spartan Autonomous Robotics Team, Michigan State University • *Aug 2026 – Present*

- Co-lead the electrical subteam (3 leads, ~12-person executive board) on the team's primary quadruped robot platform; own schematic capture and ECAD, power distribution, wiring harness design, sensor integration, and firmware bring-up.
- Establishing the electrical source of truth for the team: component registry, pinout and harness documentation, power budget, and bring-up / smoke-test procedures.
- Designing a beginner electrical training track for new members covering power/ground/signal fundamentals, breadboarding, sensors and actuators, battery safety, and first power-on checks.
- Specify electrical BOMs for recruit mini-project kits and coordinate with the Operations lead on inventory gaps and purchase requests.

TECHNICAL PROJECTS

Closed-Loop Analog Motor Control System — ECE 303: Electronics Laboratory • *Fall 2025*

- Designed a three-stage closed-loop analog DC motor controller; generated a 5 kHz PWM signal with a 555 timer and 20–80% adjustable duty cycle using independent charge/discharge paths and 1N4148 diodes for asymmetric RC timing.
- Implemented an RC low-pass filter (3.9 Ω, 150 μF) to convert the PWM output to an 800 mV DC reference; selected component values from RC transient and voltage-divider analysis to hold under 2% error from target.
- Built a closed-loop current regulation stage around a 741 op-amp driving a complementary TIP31/TIP32 push-pull pair with a 10 Ω shunt-resistor feedback network; hand-calculated all R/C values from first principles and verified performance on oscilloscope, DMM, and benchtop supply.

STM32 Environmental Monitor — Personal Project (C, STM32CubeIDE/CubeMX) • *github.com/rafayetm0rtaza*

- Built a real-time indoor environmental monitor on an STM32F411RE Nucleo board (Arm Cortex-M4, 100 MHz).
- Integrated an MQ135 gas sensor through the 12-bit ADC with CO_2 ppm conversion, a DHT22 temperature/humidity sensor over a 1-wire protocol, and an SSD1306 OLED display over I2C1.
- Configured peripheral pin mapping (ADC1, I2C1, GPIO), the clock tree for 100 MHz operation, and alternate-function routing in STM32CubeMX.

Bare-Metal Arm Cortex-M0+ Programming — ECE 331: Microprocessors & Digital Systems • *Spring 2026*

- Programmed an FRDM-KL25Z in Arm Thumb assembly; configured GPIO through direct register manipulation (SIM_SCGC5, PCR MUX, PDDR, PSOR/PCOR/PTOR) for RGB LED control.
- Implemented NVIC interrupt handling for button inputs (priority, enable registers, ISR-driven state transitions); debugged using Keil μVision memory windows and little-endian byte inspection.

8-Bit Arithmetic Logic Unit in HDL — ECE 331, AMD Xilinx Vivado • *Spring 2026*

- Designed and implemented an 8-bit ALU in HDL, covering RTL synthesis, functional simulation, and the FPGA implementation workflow.

Additional Projects

- **ShelterMEON** (CalHacks 11.0, UC Berkeley) — full-stack emergency shelter-finder web app in React and Node.js/Express with voice-to-text search and Google Maps visualization.
- **Personal portfolio site** — interactive Tesla coil visualization built with Three.js, deployed on GitHub Pages.

TECHNICAL SKILLS

Programming & HDL: C, Embedded C, Python, MATLAB, JavaScript; Verilog, VHDL
EDA & Simulation: LTspice, KiCad (schematic capture, PCB layout), AMD Xilinx Vivado, STM32CubeIDE/CubeMX, Keil μVision
Analog & Circuit Design: op-amp circuits, BJT push-pull output stages, 555-timer PWM generation, RC filter design, biasing and small-signal analysis, ADC/DAC interfacing
Embedded & Interfaces: Arm Cortex-M0+/M4, register-level GPIO, NVIC interrupts, I2C, SPI, UART, on-chip ADC
Lab Instruments: oscilloscope, function generator, benchtop power supply, digital multimeter
Software Tools: Git, GitHub, Visual Studio Code

EXPERIENCE

Student Clerical Assistant — Brody/West Circle Residence Halls, Michigan State University • *Nov 2022 – Present*

- Process incoming mail, maintain records, and reconcile student accounts for a residence community of several hundred.

Student Research Assistant — University Outreach and Engagement, Michigan State University • *May – Sep 2025*

- Processed statewide after-school program survey data using Remark and SPSS; prepared stakeholder presentations on program impact.

Data Analysis Intern — HEXA (EdTech startup) • *Jun – Aug 2022*

- Analyzed learner engagement and content performance data in Excel to support curriculum decisions and founding-team reporting.

INVOLVEMENT & AWARDS

Member, IEEE Student Branch, Michigan State University • Duke of Edinburgh's Award (Bronze, Silver) • Regional Mathematics Olympiad Award • Volunteer educator, underserved community school